

Philip-Chandan Brain Imaging Pipeline

UCSF longitudinal glioma: Phase 0 spike, expert segmentation, and clinical napari QC

QBIHack brain-cancer-sim | Generated 2026-06-28

Philip-Chandan owns the brain radiomics pipeline: download real glioma MRI with expert segmentations (not Otsu heuristics), validate NiftI pairs, export raw 3D numpy arrays, and hand volumes to Vinesh for PDE growth simulation. Phase 0 locked UCSF-ALPTDG patient 100002 baseline as the integration spike; Vinesh's prepare_pde_input.py now resamples the expert mask into a 64^3 solver grid and solve_growth() runs end-to-end on the spike slug.

1. Phase 0 spike patient

Field	Value
Dataset	UCSF Longitudinal Glioma
Patient ID	100002
Diagnosis	Oligodendroglioma, WHO grade 2
IDH / MGMT	mut / positive
Timepoint	baseline (time1)
Slug	glioma_ucsf_100002_baseline

Unlike breast-cancer-sim (TCIA DICOM + Otsu), brain datasets ship NiftI volumes with radiologist-drawn tumor masks. We use those masks as ground truth for QC and for Vinesh's PDE initial condition per handoff_contract.json v1.0.0.

2. Pipeline modules (Phase 0)

Module	Role
nifti_extractor.py	Load NiftI as (Z,Y,X) float32; resolve UCSF paths; validate MR+mask pairs
export_raw_extract.py	Write raw .npy + JSON sidecar; copy expert mask to segmentations/
qc_slice_plot.py	Mid-Z PNG with lime expert contour for visual QC
view_volume_napari.py	Interactive 3D QC viewer with clinical display controls
cohort/cohort_discovery.py	Scan local NiftI trees; find longitudinal cases; audit cohort.json
download_mu_glioma_post.py	TCIA Faspex helper for MU-Glioma-Post backup dataset
../vinesh/prepare_pde_input.py	Resample expert mask, normalize MR, crop to 64^3 PDE input
qc_pde_plot.py	QC PNGs for resampled expert mask + cropped PDE volume slices

On-disk outputs

Artifact	Path
Raw MR volume	data/processed/raw-extract-philip-chandan/glioma_ucsf_100002_baseline.npy
Metadata sidecar	.../glioma_ucsf_100002_baseline.json
Expert mask	data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz
QC overlay PNG	data/qc/slice-plots-philip-chandan/glioma_ucsf_100002_baseline_mid-z-overlay.png
PDE prep QC	data/qc/pde-prep-vinesh/glioma_ucsf_100002_baseline_pde-input-mid-z.png
PDE input (Vinesh)	data/processed/pde-input-vinesh/glioma_ucsf_100002_baseline.npy

Spike volume metadata

Property	Value
Shape (Z,Y,X)	[155, 240, 240]
Spacing (mm)	[1.0, 1.0, 1.0]
Source MR	data/raw/ucsf_alptdg/100002/100002_time1_t1ce.nii.gz
Tumor voxels	8,137 (expert mask, full volume)

PDE input metadata (Vinesh)

Property	Value
Shape (Z,Y,X)	[64, 64, 64]
Spacing (mm)	[1.0, 1.0, 1.0]
Segmentation	expert_mask
Tumor voxels	8,137 (cropped 64^3 grid)
Source mask	data/processed/segmentations/glioma_ucsf_100002_baseline_mask.nii.gz

3. Visual QC - expert segmentation overlay

qc_slice_plot.py picks the axial slice with the most tumor voxels and draws a lime contour from the UCSF expert mask. This confirms mask alignment before export and before handing data to Vinesh.

glioma_ucsf_100002_baseline z=81

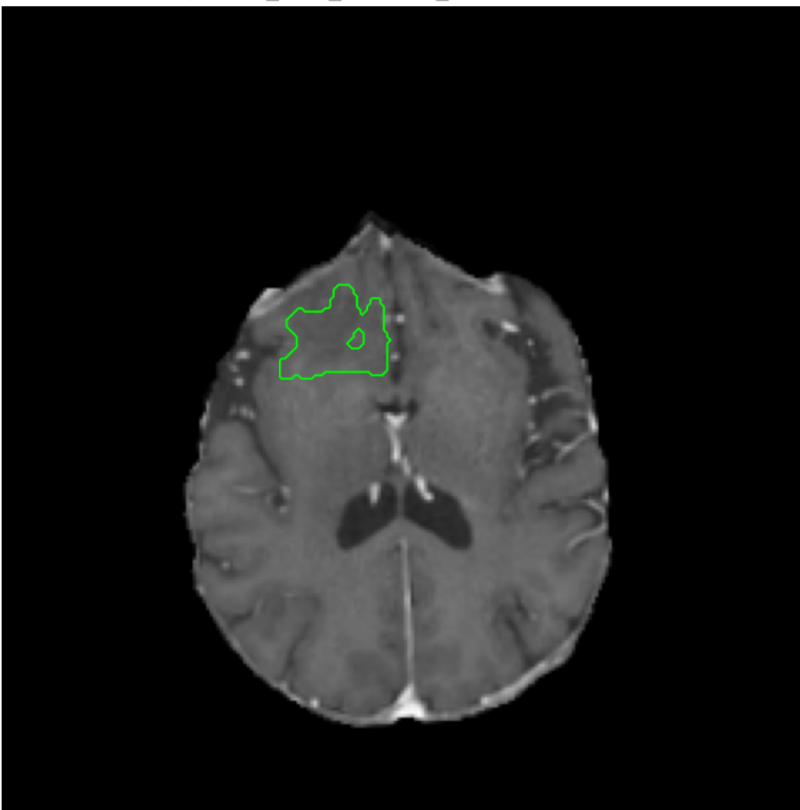


Figure 1. UCSF patient 100002 baseline T1ce with expert segmentation contour (lime). Slice chosen at maximum tumor cross-section.

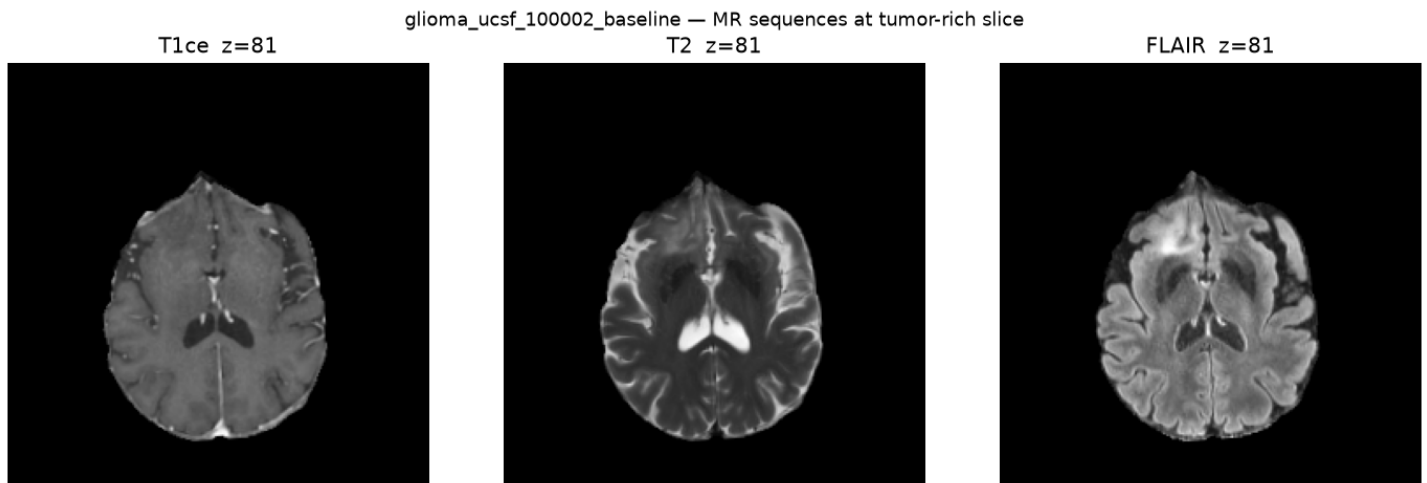


Figure 2. Same z slice across MR sequences available for this timepoint. Low-grade gliomas may show subtle T1ce enhancement; FLAIR often highlights edema.

4. MRI sequences - what T1, T2, FLAIR, and T1ce mean

Hospital brain MRI protocols acquire several contrasts of the same anatomy. Radiologists read them together; our napari viewer loads all UCSF series for the spike timepoint so you can toggle layers like on a PACS workstation.

Sequence	Full name	What you see	Glioma relevance
T1	T1-weighted	Anatomy; CSF dark, fat bright	Baseline anatomy; pre-contrast reference
T1ce	T1 post-contrast (T1+C)	Areas with leaky blood vessels brighten after gadolinium	Enhancing tumor rim; spike export uses T1ce as primary MR
T2	T2-weighted	Fluid bright; good soft-tissue contrast	Tumor and edema often appear brighter than normal brain
FLAIR	Fluid-attenuated inversion recovery	T2-like but CSF suppressed (dark ventricles)	Peritumoral edema and infiltrative signal stand out clearly

For our grade 2 IDH-mut oligodendroglioma spike, T1ce enhancement is subtle - mean intensity inside the expert mask is almost identical to surrounding brain. FLAIR is often more informative for edema. The expert mask delineates the lesion using all available clinical context, not brightness on a single sequence.

5. Clinical napari viewer - view_volume_napari.py

We extended the napari QC viewer beyond a simple grayscale overlay. Controls live in the right dock panels. CLAHE and window/level adjustments are labeled QC-only - they help engineers inspect data, not replace diagnostic PACS reading.

Window / Level (WW/WL)

Radiology displays map MR intensities to gray using a window (width) and level (center). Narrow windows increase contrast but clip bright/dark regions. Our defaults compute level and width from brain voxels only (2nd-98th percentile), ignoring black padding outside the head. Sliders update all MR layers; hold Ctrl and drag on the canvas for click-drag WW/WL like a clinical viewer.

Multi-sequence layers

For UCSF patient folders, the viewer auto-loads T1, T1ce, T2, and FLAIR when NiftI files exist. T1ce is visible by default with the expert overlay. Toggle FLAIR or T2 in napari's layer list to compare edema vs enhancement on the same slice.

Plane selector and orthogonal MPR grid

The Plane dropdown switches axial, coronal, or sagittal viewing on the main canvas while keeping the segmentation aligned. Orthogonal MPR grid opens a 1x3 linked layout (axial | coronal | sagittal) for navigating the tumor in three planes; the overlay is hidden in grid mode because napari assigns one layer per grid cell.

CLAHE (Contrast Limited Adaptive Histogram Equalization)

CLAHE enhances local contrast slice-by-slice inside the brain mask. It can make subtle texture differences easier to see during QC. It is not a physical imaging sequence and must not be used for diagnosis or as PDE input - it is a display filter only, clearly marked in the viewer.

Overlay toggle

Hide overlay / Show overlay toggles the expert segmentation labels so you can inspect underlying MR intensity. For subtle T1ce lesions, pairing this with FLAIR and WW/WL adjustment is the intended workflow.

Launch command

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/view_volume_napari.py --slug glioma_ucsf_100002_baseline
```

6. Expert-mask PDE prep - prepare_pde_input.py (Vinesh)

After Philip-Chandan raw export, Vinesh prepares PDE-ready volumes in vinesh/prepare_pde_input.py per handoff_contract.json v1.0.0. Unlike breast-cancer-sim (Otsu on DCE-MRI), brain v1 uses the dataset expert mask - no Otsu fallback.

- Resample MR (linear) and expert mask (nearest) to isotropic 1 mm spacing
- Min-max normalize MR intensities to [0, 1]
- Keep continuous normalized density inside the expert mask; zero background
- Crop/pad to max 64^3 centered on mask center of mass

Continuous tumor values (not a binary mask) are required so the PDE logistic growth term $\rho \cdot u \cdot (1-u)$ can evolve. qc_pde_plot.py documents the same pipeline via prepare_pde_stages() for the PDF figures below.

100002 expert mask z=81 (mask vox=8137)

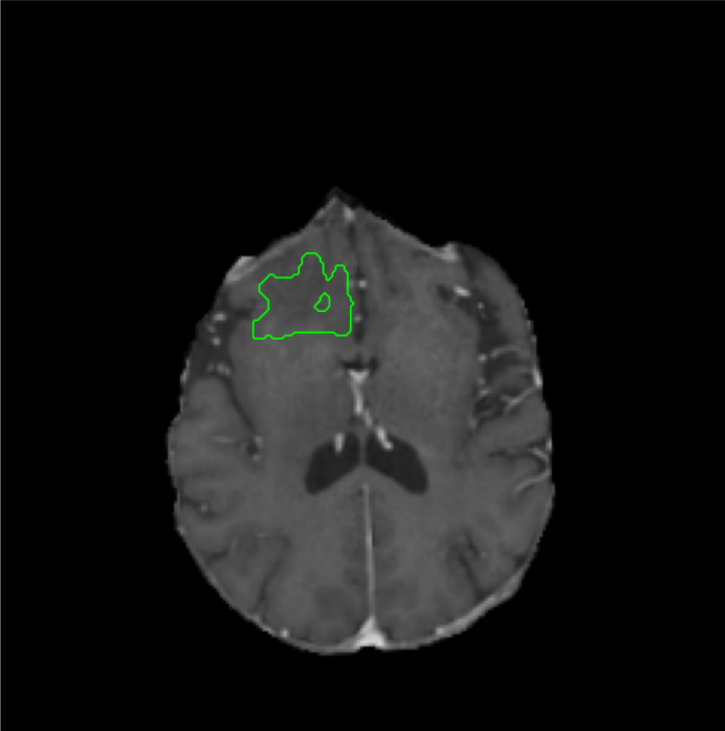


Figure 3. UCSF patient 100002 after resample + normalize. Lime contour is the expert tumor mask on the 1 mm grid before crop; background will be zeroed.

100002 PDE input z=30 shape=[64, 64, 64]

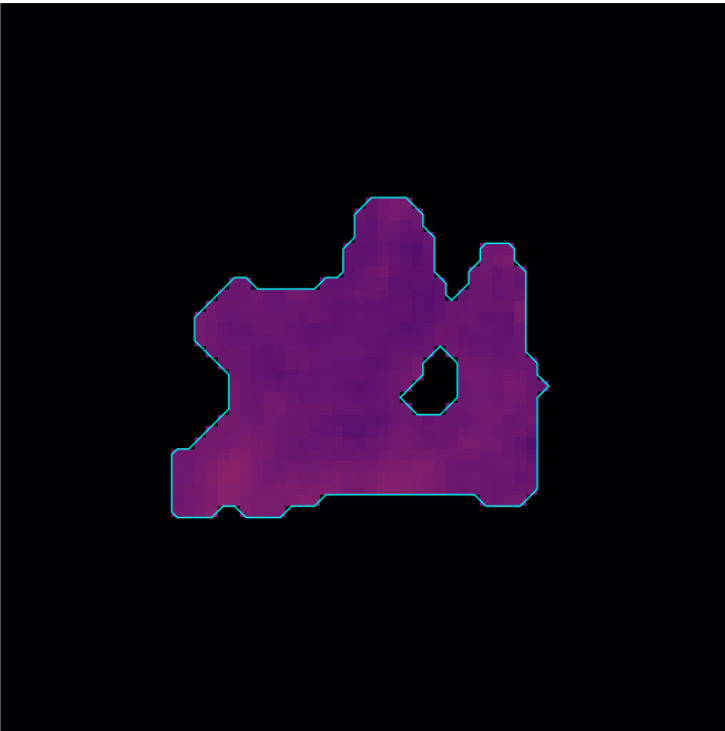


Figure 4. Cropped PDE input (64^3 , 1 mm). Inferno colormap shows continuous initial tumor burden; cyan contour marks voxels > 0 .

Solver smoke test on spike PDE input

run_growth() on glioma_ucsf_100002_baseline: 51 frames, shape [64, 64, 64], tumor volume ($u > 0$) at $t=0$ = 262144 mm^3 , at $t=\text{end}$ ($u \geq 0.5$) = 5738 mm^3 . Growth term is active because initial burden is continuous inside the expert ROI.

Regenerate PDE artifacts

```
cd brain-cancer-sim
.venv/bin/python simulation-vinesh-philip-chandan/vinesh/prepare_pde_input.py
.venv/bin/python simulation-vinesh-philip-chandan/philip-chandan/qc_pde_plot.py
```

7. Handoff contract summary

Philip-Chandan (raw extract)	Vinesh (PDE input)
Raw MR intensities, not normalized	Resample/crop toward max 64^3 , 1 mm spacing
spacing_mm in JSON sidecar	Normalize to [0, 1] inside tumor region
Expert mask path in metadata	Expert mask resampled - no Otsu fallback
Axis order (Z, Y, X)	solve_growth() initial condition

8. Phase 0 summary and next steps

Phase 0 delivered a working NIfTI ingest path, UCSF spike export, expert-mask QC figures, a clinical-style napari viewer, Vinesh PDE prep from the expert mask into 64^3 continuous-density input, and a successful solve_growth() smoke test on patient 100002 baseline. Next: export follow-up timepoint (100002_time2), add a second patient or grade for the demo toggle, and publish manifest.json v1.0.0.

Deliverables completed

Stage	Deliverable	Status
Cohort	cohort.json rev1 + COHORT.md	UCSF spike locked
Download	UCSF NIfTI under data/raw/ucsf_alptdg/100002/	On disk
Extract	nifti_extractor.py + tests	Passing
Export	Raw .npy + .json + mask (glioma_ucsf_100002_baseline)	Done
QC	Slice overlay PNG + napari viewer	Done
PDE prep	pde-input-vinesh/glioma_ucsf_100002_baseline	Done
Solver	solve_growth() on spike PDE input	Smoke test passed